

# Business Engineering – Exercise Sheet: Diagonalization, Matrix Powers, and Long-Run Logistics Dynamics

Bachelor in Business Engineering

3 June 2026

## Exercises on Diagonalization, Symmetric and Non-Symmetric Matrices, and Dominant Logistics Modes

Use exact expressions whenever convenient and round numerical answers to three decimals when needed. Unless otherwise stated, vectors are column vectors and matrices act from the left:

$$x_{k+1} = Ax_k.$$

In logistics interpretations, use the convention that columns represent source objects and rows represent target objects unless explicitly stated otherwise. Thus, the entry  $a_{ij}$  describes the fraction, cost, weight, travel relation, or operational influence from source  $j$  to target  $i$ .

When useful, use

$$\begin{aligned} Av &= \lambda v, & w^T A &= \lambda w^T, & A^T w &= \lambda w, \\ \det(A - \lambda I) &= 0, & \chi_A(\lambda) &= \lambda^2 - \operatorname{tr}(A)\lambda + \det(A) & \text{ for } A \in \mathbb{R}^{2 \times 2}, \\ A &= PDP^{-1}, & A^k &= PD^kP^{-1}, & D^k &= \begin{pmatrix} \lambda_1^k & 0 \\ 0 & \lambda_2^k \end{pmatrix} & \text{ for } D = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}. \end{aligned}$$

For a column-stochastic redistribution matrix, the column sums are equal to 1, and total stock is conserved when the process is read as  $x_{k+1} = Ax_k$ .

### Part A – Diagonalization, powers, and long-run logistics behavior (Exercises 1–10)

#### 1. Ex 1: Building the diagonalization from logistics modes

A two-hub redistribution process has two independent right eigenvectors

$$v_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix},$$

with corresponding eigenvalues

$$\lambda_1 = 1, \quad \lambda_2 = 0.4.$$

- (a) Explain in one logistics sentence what it means that  $v_1$  is preserved in direction under the redistribution matrix. (4 Points)
- (b) Construct the matrix  $P$  whose columns are the eigenvectors  $v_1$  and  $v_2$ . (4 Points)
- (c) Construct the diagonal matrix  $D$  whose diagonal entries are the corresponding eigenvalues. (4 Points)
- (d) Compute  $\det(P)$ . Explain why  $\det(P) \neq 0$  is necessary for diagonalization. (4 Points)
- (e) Compute  $P^{-1}$ . (4 Points)

2. **Ex 2: Constructing a redistribution matrix from  $PDP^{-1}$**

Use

$$P = \begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix}, \quad D = \begin{pmatrix} 1 & 0 \\ 0 & 0.4 \end{pmatrix}.$$

- (a) Compute

$$A = PDP^{-1}.$$

(8 Points)

- (b) Verify directly that

$$Av_1 = 1 \cdot v_1, \quad Av_2 = 0.4 \cdot v_2.$$

(4 Points)

- (c) Check whether the columns of  $A$  sum to 1. Interpret the result operationally. (4 Points)
- (d) Identify the dominant eigenvalue and dominant eigenvector. What long-run hub ratio do they imply? (4 Points)

3. **Ex 3: Decomposing an initial stock vector into eigenvectors**

Use the matrix  $A$  from Exercise 2. Suppose the initial stock vector is

$$x_0 = \begin{pmatrix} 120 \\ 0 \end{pmatrix}.$$

- (a) Find scalars  $\alpha, \beta$  such that

$$x_0 = \alpha v_1 + \beta v_2.$$

(6 Points)

- (b) Write

$$x_k = A^k x_0$$

in modal form using  $\alpha, \beta, \lambda_1, \lambda_2, v_1, v_2$ . (5 Points)

- (c) Derive an explicit coordinate formula for  $x_k$ . (5 Points)

(d) Compute  $x_1$  and  $x_2$ . (2 Points)

(e) Compute the limiting stock vector as  $k \rightarrow \infty$ . (2 Points)

**4. Ex 4: Same total stock, different initial imbalance**

Again use

$$A = \begin{pmatrix} 0.6 & 0.2 \\ 0.4 & 0.8 \end{pmatrix}, \quad v_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Consider two initial stock vectors:

$$x_0^{(a)} = \begin{pmatrix} 60 \\ 60 \end{pmatrix}, \quad x_0^{(b)} = \begin{pmatrix} 0 \\ 120 \end{pmatrix}.$$

(a) Decompose  $x_0^{(a)}$  into the eigenvector basis  $\{v_1, v_2\}$ . (5 Points)

(b) Decompose  $x_0^{(b)}$  into the eigenvector basis  $\{v_1, v_2\}$ . (5 Points)

(c) Derive  $A^k x_0^{(a)}$  and  $A^k x_0^{(b)}$ . (5 Points)

(d) Show that both initial vectors converge to the same long-run stock vector. Explain why this is operationally reasonable. (3 Points)

(e) Determine the smallest integer  $k$  such that  $0.4^k < 0.01$ . Interpret this as decay of the imbalance mode. (2 Points)

**5. Ex 5: Comparing slow and fast redistribution designs**

Two possible redistribution designs are

$$A_{\text{slow}} = \begin{pmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{pmatrix}, \quad A_{\text{fast}} = \begin{pmatrix} 0.55 & 0.30 \\ 0.45 & 0.70 \end{pmatrix}.$$

Both act by

$$x_{k+1} = Ax_k.$$

(a) Verify that both matrices are column-stochastic. (4 Points)

(b) Compute the eigenvalues of  $A_{\text{slow}}$ . (4 Points)

(c) Compute the eigenvalues of  $A_{\text{fast}}$ . (4 Points)

(d) Compute one right eigenvector for  $\lambda = 1$  for each matrix. Compare the long-run stock ratios. (4 Points)

(e) Which design removes initial imbalance faster? Justify your answer using the magnitude of the second eigenvalue. (2 Points)

(f) Explain briefly why faster redistribution is not automatically better in logistics. Mention one possible trade-off. (2 Points)

**6. Ex 6: Computing  $A^5x_0$  using diagonalization**

A two-location logistics system has right eigenvectors

$$v_1 = \begin{pmatrix} 3 \\ 2 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix},$$

with eigenvalues

$$\lambda_1 = 1, \quad \lambda_2 = 0.6.$$

Thus

$$P = \begin{pmatrix} 3 & 1 \\ 2 & -1 \end{pmatrix}, \quad D = \begin{pmatrix} 1 & 0 \\ 0 & 0.6 \end{pmatrix}.$$

The initial stock vector is

$$x_0 = \begin{pmatrix} 100 \\ 20 \end{pmatrix}.$$

(a) Compute  $P^{-1}$ . (4 Points)

(b) Decompose  $x_0$  as

$$x_0 = \alpha v_1 + \beta v_2.$$

(5 Points)

(c) Write the general expression for

$$x_k = A^k x_0.$$

(5 Points)

(d) Compute  $x_5$ . Round to three decimals. (4 Points)

(e) Compute the limiting stock vector as  $k \rightarrow \infty$ . (2 Points)

**7. Ex 7: Modal decomposition of a three-region logistics system**

A logistics update matrix  $A$  has three independent right eigenvectors

$$v_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \quad v_3 = \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix},$$

with eigenvalues

$$\lambda_1 = 1, \quad \lambda_2 = 0.5, \quad \lambda_3 = 0.25.$$

The initial stock vector is given in modal form:

$$x_0 = 90v_1 + 30v_2 - 10v_3.$$

(a) Compute  $x_0$  in ordinary coordinates. (4 Points)

(b) Write the general expression for

$$x_k = A^k x_0$$

using the modal decomposition. (5 Points)

- (c) Compute  $x_1$ . (3 Points)
- (d) Compute  $x_2$ . (4 Points)
- (e) Identify which mode persists, which mode decays faster, and what this means for regional stock balancing. (4 Points)

8. **Ex 8: Dominant growth mode in a demand-amplification process**

A demand-pattern update has eigenvectors

$$v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix},$$

with eigenvalues

$$\lambda_1 = 1.2, \quad \lambda_2 = 0.7.$$

The initial demand pattern is

$$x_0 = \begin{pmatrix} 100 \\ 60 \end{pmatrix}.$$

- (a) Decompose  $x_0$  as

$$x_0 = \alpha v_1 + \beta v_2.$$

(5 Points)

- (b) Derive the formula for  $x_k$ . (5 Points)
- (c) Compute  $x_3$ . Round to three decimals. (4 Points)
- (d) Which mode dominates as  $k$  becomes large? Explain using the eigenvalues. (3 Points)
- (e) Interpret the difference between a growing common demand mode and a decaying imbalance mode. (3 Points)

9. **Ex 9: Negative eigenvalue and alternating imbalance**

Consider the redistribution matrix

$$A = \begin{pmatrix} 0.25 & 0.75 \\ 0.75 & 0.25 \end{pmatrix}.$$

It has eigenvectors

$$v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

- (a) Compute the eigenvalues corresponding to  $v_1$  and  $v_2$ . (5 Points)
- (b) Verify that  $A$  is column-stochastic. (3 Points)

(c) For

$$x_0 = \begin{pmatrix} 100 \\ 0 \end{pmatrix},$$

decompose  $x_0$  into the eigenvector basis. (4 Points)

(d) Derive  $x_k = A^k x_0$ . (4 Points)

(e) Compute  $x_1, x_2, x_3$ . Explain why the imbalance changes sign from period to period. (4 Points)

**10. Ex 10: Reading a repeated process from its modal formula**

For a three-location logistics process, the state after  $k$  periods is known to be

$$x_k = 50 \cdot 1^k \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + 20 \cdot 0.8^k \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} + 10 \cdot 0.3^k \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix}.$$

(a) Compute the initial state  $x_0$ . (4 Points)

(b) Compute  $x_1$ . (4 Points)

(c) Compute  $x_3$ . Round to three decimals. (4 Points)

(d) Identify the persistent mode, the slow-decay mode, and the fast-decay mode. (4 Points)

(e) Explain in logistics language why the coefficient multiplying  $0.8^k$  remains relevant for more periods than the coefficient multiplying  $0.3^k$ . (4 Points)

**Part B – Symmetric matrices, non-symmetric matrices, left/right eigenvectors, and non-diagonalizable cases (Exercises 11–20)**

**11. Ex 11: Symmetric matrix and orthogonal eigenvectors**

Consider the symmetric matrix

$$S = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

(a) Verify that  $S = S^T$ . (2 Points)

(b) Compute the characteristic polynomial

$$\det(S - \lambda I).$$

(5 Points)

(c) Determine the eigenvalues of  $S$ . (4 Points)

(d) Compute one eigenvector for each eigenvalue. (5 Points)

- (e) Show that the two eigenvectors are orthogonal. Explain why this is typical for real symmetric matrices. (4 Points)

**12. Ex 12: Orthogonal diagonalization**

Consider

$$S = \begin{pmatrix} 5 & 2 \\ 2 & 5 \end{pmatrix}.$$

- (a) Compute the eigenvalues of  $S$ . (4 Points)
- (b) Compute one eigenvector for each eigenvalue. (4 Points)
- (c) Normalize the eigenvectors and build an orthogonal matrix  $Q$ . (4 Points)
- (d) Construct the diagonal matrix  $D$  such that

$$S = QDQ^T.$$

(4 Points)

- (e) Verify that  $Q^T Q = I$ . Explain why  $Q^{-1} = Q^T$  makes symmetric matrices especially convenient. (4 Points)

**13. Ex 13: Non-symmetric road-time matrix**

In this exercise only, let  $T_{ij}$  denote the travel time from origin  $i$  to destination  $j$ . Consider

$$T = \begin{pmatrix} 0 & 14 & 22 \\ 17 & 0 & 12 \\ 20 & 16 & 0 \end{pmatrix} \text{ minutes.}$$

- (a) Compute  $T^T$ . (4 Points)
- (b) Determine whether  $T$  is symmetric. Identify every pair of locations for which the travel time depends on the travel direction. (5 Points)
- (c) Compute the round-trip matrix

$$R = T + T^T.$$

(5 Points)

- (d) Which two-location round trip is shortest? (3 Points)
- (e) Explain why Euclidean straight-line distance is symmetric, while real road travel time is often non-symmetric. (3 Points)

**14. Ex 14: Left eigenvectors and conservation of total stock**

Consider

$$A = \begin{pmatrix} 0.8 & 0.1 \\ 0.2 & 0.9 \end{pmatrix}.$$

- (a) Verify that  $A$  is column-stochastic. (3 Points)
- (b) Compute the eigenvalues of  $A$ . (5 Points)
- (c) Compute right eigenvectors for both eigenvalues. (5 Points)
- (d) Compute left eigenvectors for both eigenvalues by solving

$$A^T w = \lambda w.$$

(5 Points)

- (e) Show that

$$w = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

is a left eigenvector for  $\lambda = 1$ , and explain why it represents conservation of total stock. (2 Points)

**15. Ex 15: Source-target convention and the danger of reading a matrix in the wrong direction**

Let

$$A = \begin{pmatrix} 0.6 & 0.3 \\ 0.4 & 0.7 \end{pmatrix}.$$

Assume first that columns are sources and rows are targets, so that

$$x_{k+1} = Ax_k.$$

- (a) Check the column sums of  $A$ . What do they imply? (4 Points)
- (b) For

$$x_0 = \begin{pmatrix} 100 \\ 0 \end{pmatrix},$$

compute  $x_1 = Ax_0$  and verify that total stock is conserved. (4 Points)

- (c) Now incorrectly read the same matrix using row-vector evolution:

$$z_1^T = z_0^T A, \quad z_0^T = (100 \ 0).$$

Compute  $z_1^T$ . Is total stock conserved? (5 Points)

- (d) Compute  $A^T$ . Explain how transposition changes the source-target interpretation. (4 Points)
- (e) State one practical logistics risk caused by confusing row-wise and column-wise interpretations. (3 Points)

**16. Ex 16: Non-symmetric but diagonalizable**

Consider

$$N = \begin{pmatrix} 1 & 2 \\ 0 & 0.5 \end{pmatrix}.$$



- (a) Show that  $N \neq N^T$ . (2 Points)
- (b) Compute the eigenvalues of  $N$ . (4 Points)
- (c) Compute one right eigenvector for each eigenvalue. (5 Points)
- (d) Compute one left eigenvector for each eigenvalue by solving

$$N^T w = \lambda w.$$

(5 Points)

- (e) Are the right eigenvectors orthogonal? Explain why a non-symmetric matrix may be diagonalizable without being orthogonally diagonalizable. (4 Points)

**17. Ex 17: A matrix that is not diagonalizable**

Consider

$$J = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

- (a) Compute the characteristic polynomial of  $J$ . (4 Points)
- (b) Determine the eigenvalue of  $J$  and its algebraic multiplicity. (4 Points)
- (c) Solve
 
$$(J - I)v = 0$$
 and determine the eigenspace. (5 Points)
- (d) How many linearly independent eigenvectors does  $J$  have? (3 Points)
- (e) Explain why  $J$  is not diagonalizable. (4 Points)

**18. Ex 18: Powers of a non-diagonalizable coupling process**

Use again

$$J = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

- (a) Verify directly that

$$J^2 = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}.$$

(3 Points)

- (b) Show by induction or repeated multiplication that

$$J^k = \begin{pmatrix} 1 & 0 \\ k & 1 \end{pmatrix}.$$

(6 Points)

(c) For

$$x_0 = \begin{pmatrix} 10 \\ 5 \end{pmatrix},$$

compute  $J^k x_0$ . (4 Points)

(d) Compute  $x_1, x_3, x_{10}$ . (4 Points)

(e) Give one logistics interpretation of a repeated process where a coupling term accumulates linearly even though the only eigenvalue is 1. (3 Points)

**19. Ex 19: Practical diagonalizability criteria**

For each of the following matrices, decide whether it is diagonalizable over  $\mathbb{R}$ :

$$A_1 = \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix},$$

$$A_3 = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \quad A_4 = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}.$$

(a) Compute the eigenvalues of  $A_1, A_2, A_3, A_4$ . (8 Points)

(b) For each matrix, state whether it has distinct eigenvalues, is symmetric, or has a repeated eigenvalue. (4 Points)

(c) Decide whether each matrix is diagonalizable. Justify each decision using the practical criteria:

distinct eigenvalues  $\Rightarrow$  diagonalizable,

real symmetric matrix  $\Rightarrow$  diagonalizable,

repeated eigenvalue  $\nRightarrow$  automatically diagonalizable.

(6 Points)

(d) For the non-diagonalizable case, compute the eigenspace and explain why there are not enough independent eigenvectors. (2 Points)

**20. Ex 20: Integrated logistics design decision**

A logistics provider compares three weekly redistribution matrices:

$$B_1 = \begin{pmatrix} 0.85 & 0.10 \\ 0.15 & 0.90 \end{pmatrix}, \quad B_2 = \begin{pmatrix} 0.52 & 0.32 \\ 0.48 & 0.68 \end{pmatrix}, \quad B_3 = \begin{pmatrix} 0.20 & 0.80 \\ 0.80 & 0.20 \end{pmatrix}.$$

(a) Verify that all three matrices are column-stochastic. (3 Points)

(b) Compute the eigenvalues of  $B_1, B_2, B_3$ . (6 Points)

(c) Compute a right eigenvector for  $\lambda = 1$  for each matrix and interpret the corresponding long-run stock ratio. (5 Points)

- (d) Which matrix converges fastest to its long-run ratio? Which matrix creates an alternating imbalance? Justify using the second eigenvalue. (3 Points)
- (e) Give a short engineering recommendation. Mention at least two criteria: speed of convergence, operational stability, transport effort, robustness, service level, or risk of overcorrection. (3 Points)